Kinematics in one dimension

Displacement is the change in position of an object. It has a direction as well as a size, so it is a vector. Distance is how far the object travels along its path.

Average velocity is displacement divided by the time it takes. Speed is the size of the velocity. A car that drives 30 m east in 3 s has an average velocity of 10 m/s.

Acceleration is the rate of change of velocity. When the velocity of an object does not change, its acceleration is zero, even if it moves very fast.

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